

LOT-HW1 — Plan (Document 01 of 06)

Product: LOT Pager / LOT Computer, hardware unit 1 (LOT-HW1) **Owner:** S-2 Vadik Marmeladov · COSMO® CIA **Status:** PLAN — nothing is built or ordered. Dated 2026-10-04. **Set:** 01 Plan · 02 BOM · 03 Roadmap · 04 Firmware · 05 API connector · 06 Manufacturing & manuals

Evidence labels used throughout: **VERIFIED** (read in this repo or a cited source), **ESTIMATE** (engineering judgement, needs a quote or a bench test), **DECISION** (S-2 must choose). Prices are ESTIMATES, not quotes. Sandbox limit: lot-systems.com, brand.lot-systems.com, institute.lot-systems.com and most vendor sites were blocked in this session, so brand/CQGS pages were **not read**. Brand tokens below come from tailwind.config.js in this repo.

1. What it is

A 40 × 40 mm flat stainless-steel object that receives autonomous notifications from lot-systems.com ("Coffee time!"), shows them on a small screen, and lets the wearer press one button, **Copy**, which sends a signal back to the site's **Log** tab.

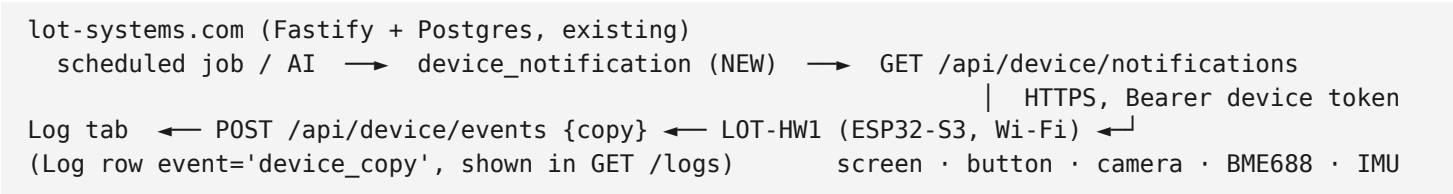
Face	Content
Side A	Polished stainless steel, plain mirror. Nothing on it.
Side B	Screen, camera, one button.

Body: two stainless parts (back plate + frame/bezel). Charging: wireless (Qi). Sensors: weather + off-the-shelf "AI-grade" parts. Run: 100 units.

2. Requirement map (your 19 points → where handled)

#	Your line	Handled in
1	PCBWay	Doc 02 (BOM), Doc 06 (order package)
2	Pager-like notification from AI site	Doc 05 §3 (notification flow)
3	2-part stainless body	§4 below, Doc 06
4	4×4 cm × 5 mm silver square	§4 — 5 mm is high-risk, see §5
5	Camera	Doc 02, Doc 04 §6
6	LOT API connector	Doc 05
7	PDF manuals	Doc 06 §4 (PDFs generated from these .md files)
8	Compress info each session	Session report + benchmark ledger/lexicon (docs/benchmark/)
9	Firmware documents	Doc 04
10	Software to connect with firmware	Doc 04 §8 (web flasher + provisioning)
11	Separate documents	This 6-document set
12, 19	Charger / wireless charger	§6, Doc 02
13	100-unit run	Doc 03 phase P4, Doc 06
14, 15	Weather + AI-grade sensors	§7, Doc 02
16	Button "Copy" → Log tab	Doc 05 §4
17, 18	Side A polished / Side B screen+camera+button	§1, §4

3. Architecture



4. Mechanical concept

- **Body:** 316L austenitic stainless (non-magnetic, corrosion-resistant) ESTIMATE-choice; 304 acceptable. Two parts: (1) back plate, mirror-polished; (2) frame + front bezel with window for screen, camera aperture, button hole. Joined by laser weld or 4 micro-screws + gasket (DECISION: weld is cleaner, screws are serviceable).
- **Footprint:** 40 × 40 mm, corner radius ~3 mm (DECISION).
- **Front:** thin cover glass over screen; small separate lens window for camera; one tactile button (side-mounted tact, or a pressed bezel region).
- **Brand colours for the screen UI** (from this repo's `tailwind.config.js`, VERIFIED): background #1a1a1a / #050505, accent blue #43aff3, yellow #fef17b, text white; font Arial/Helvetica. Re-check against brand.lot-systems.com before design freeze (not readable in this session).

5. The three physics problems (read these first)

These are the reasons a naive build fails. Each has a mitigation and a bench test in the roadmap.

5.1 Thickness: 5 mm is very hard

Stack budget (ESTIMATE, mm):

Layer	Track A "5.0 target"	Track B "realistic Gen-1"
Steel back plate	0.5	0.8
Qi coil + ferrite	0.3	0.5
Battery (ultra-thin LiPo)	1.2	2.0
PCB + tallest part (BME688 0.9 mm, bare QFN 0.85 mm)	1.4	1.8
Display module (glass)	1.2	1.5
Cover glass / bezel lip	0.4	0.6
Total	5.0	7.2

- ESP32-S3-MINI-1 module is 15.4 × 20.5 × **2.4** mm (VERIFIED, [Espressif datasheet](#)) — too tall for Track A. Track A needs a **chip-down** ESP32-S3 design (own RF layout) which loses the module's pre-certification (see 5.4).
- Camera modules are typically 3–5 mm tall (ESTIMATE) and are the real thickness limiter; a 5 mm body forces a very thin fixed-focus module or a sensor-on-flex design.
- **Recommendation (DECISION for S-2):** build Gen-1 (EVT/DVT) at **≈7 mm using the module**, prove everything works, then chase 5 mm in a Gen-2 respin. Do not promise 5 mm to anyone for the first 100 units unless a thin camera + thin cell are confirmed in P1.

5.2 Wireless charging through steel

Qi power does not pass through a continuous stainless plate efficiently (eddy-current loss; ESTIMATE: large loss, must be measured). Options: - **A (recommended to test first): charge face-down through the front.** Coil sits behind the cover-glass side, so the polished back faces up. Needs the coil to avoid the display's metal; test coil-under-display vs coil-in-border. - **B:** small non-metal (ceramic/PEEK) disc inset in the back plate — breaks the "pure mirror" (DECISION). - **C:** very thin (≈ 0.3 mm) 316L over a ferrite-backed coil — marginal; only if bench test shows acceptable efficiency. - Receiver: TI BQ51013B-class Qi receiver (see [TI product page](#); Adafruit-style modules use it per search results) → Li-ion charger IC. Qi **logo** certification is out of scope for 100 units; interoperability test with 3 common pads instead. - Expected output is 5 V class, ≤ 1 A, which is plenty for a ~ 100 mAh cell.

5.3 Radio and sensors inside a steel box

- **Wi-Fi/BLE antenna** cannot sit inside a sealed metal can. Put the antenna at the front/glass side, ≥ 5 mm from steel (ESTIMATE keep-out), or cut a plastic-filled slot in the frame. Needs an over-the-air range test on a steel mock-up in P1. This is the biggest functional risk after thickness.
- **Weather sensor** needs airflow. A sealed body reads its own heat. Add a ~ 0.8 mm vent with hydrophobic membrane near the sensor (DECISION: breaks water-sealing claim; fine for splash only). Also expect self-heating offset; calibrate in firmware, and use the site's server-side `/api/weather` (VERIFIED route exists in `src/server/routes/api.ts`) as fallback.

5.4 Regulatory

- An unlicensed 2.4 GHz radio sold or distributed needs FCC (US, "Made in the USA" per `docs/README.md`) and CE-RED if shipped to EU. Using a pre-certified module (ESP32-S3-MINI-1) can reduce cost; chip-down means full intentional-radiator testing. Budget ESTIMATE \$5k–15k (Doc 03). Lithium cell needs UN38.3 for shipping. Confirm with a test lab before P4.
- Camera on a worn device: privacy by design — see Doc 04 §6 (capture only on button, visible indicator, no always-on stream).

6. Power

- Cell: ~ 50 – 120 mAh thin LiPo with protection (thickness drives capacity; see Doc 02 note).
- Charge path: Qi RX → charger/power-path IC → 3.3 V buck/LDO. Fuel gauge optional.
- Duty cycle (ESTIMATE): ESP32-S3 deep sleep ~ 10 – 20 μ A; wake every 1–5 min to poll over Wi-Fi (~ 100 – 300 mA for ~ 1 – 3 s); screen on only after a new notification or button press. Rough battery life ESTIMATE 1–3 days at 100 mAh with 2-min polling; to be measured in P2. A latch-style e-paper alternative is noted in Doc 03 as a Gen-2 option.

7. Sensors (weather + AI-grade, all off-the-shelf)

- **Bosch BME688:** gas + humidity + pressure + temperature, $3.0 \times 3.0 \times 0.9$ mm, on-sensor AI gas scanning (VERIFIED: [Bosch product page](#), [datasheet](#)). Covers weather (14) and an AI-grade sensor (15) in one part.
- **Bosch BMI270** 6-axis IMU (tap/wrist/motion; on-chip gesture features) — ESTIMATE-choice, part number to confirm in P1.

- **Ambient light sensor** (auto brightness; sunrise/sunset tie-in with LOT time palette) — part TBD in P1.
- Microphone: **excluded in Gen-1** (privacy, thickness).

8. Open decisions for S-2

1. Accept ≈ 7 mm Gen-1 (recommended) or hold 5 mm?
2. Charging face: back-through-glass (A) vs ceramic inset (B)?
3. Weld vs screws? Vent allowed?
4. What does the camera *do* for the user (snap-to-Log? scan?). Plan assumes **snap** → **upload** → **Log entry**; confirm.
5. Screen type: colour IPS (cheap, $\sim 1.3''$) vs AMOLED (thin, pricey) vs e-paper (low power, slow).
6. Target regions for certification (US only / US+EU).
7. Who is the first-100 audience (staff, Usership/Onyx tags per docs/ — not verified)?

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LOT-HW1 — Components Buying List (Document 02 of 06)

Evidence labels: VERIFIED = part/spec confirmed from a cited page in the 2026-10-04 research pass. ESTIMATE = judgement or price not quoted. Links marked *unverified* could not be opened from the sandbox (vendor sites blocked); open and confirm before ordering. **Machine-readable copy:** LOT-HW1-B0M.csv (same rows; import to PCBWay quote form after MPN confirmation).

Prices are rough USD per unit at ~100-unit volume, **ESTIMATE ±40%, not quotes**. PCBWay assembly is manually quoted (1–2 business days per research), so get a real quote in phase P2.

1. Electronic parts (per unit)

Ref	Function	Part (first choice)	Alt	Key spec	Est. \$	Source / link
U1	MCU + Wi-Fi/BLE	ESP32-S3-MINI-1-N8 (Gen-1, module)	ESP32-S3 chip-down (Gen-2, thinner)	15.4×20.5×2.4 mm, 8 MB flash, PCB antenna, VERIFIED	3–4	Espressif datasheet (VERIFIED); atomic14 module notes
DS1	Screen	1.3" 240×240 IPS, SPI, ST7789-class	AMOLED round/square; e-paper	Module outlines seen 26.2×29.2×1.5 mm (VERIFIED, one vendor) — pick a part ≤ 1.5 mm	5–8	Display Module 1.3" IPS ; Adafruit 4520 (reference part for prototyping)
CAM1	Camera	OV2640 2 MP DVP (JPEG on-chip)	GC2145 2 MP DVP	24-pin FPC; thickness 3–5 mm ESTIMATE → thickness risk	3–6	Arducam OV2640 mini (prototype); GC2145 RobotShop pack
U2	Weather + AI gas	Bosch BME688	BME280 (no gas)	3.0×3.0×0.9 mm, VERIFIED	8–10	Bosch ; datasheet
U3	IMU / gestures	Bosch BMI270	BMA400	LGA-14 2.5×3.0×0.83 mm, VERIFIED	2–3	Bosch BMI270 ; LCSC C2836813
U4	Ambient light	VEML7700 or equivalent	—	tiny I ² C ALS; part TBD in P1	1	<i>unverified</i> — pick in P1
U5	Qi receiver	TI BQ51013B (5 W class)	cheaper Qi RX IC in P2	Qi v1.x, ≤5 W; used by Adafruit AF-1901 module (VERIFIED via search)	2–3	TI BQ51013B <i>unverified</i> ; Adafruit datasheet
L1	Qi RX coil + ferrite	20–25 mm flex coil with ferrite sheet, ≤0.5 mm	module-integrated coil	thin type; must be bench-tested through chosen stack	2–4	vendor TBD (Würth/TDK/generic) — <i>unverified</i>
U6	Battery charger	TI BQ25185 (1 A linear, power-path)	BQ25101 (250 mA, 1.4×1.0×0.5 mm, VERIFIED)	BQ25101 is smaller and enough for ~100 mAh cell	1–2	BQ25185 ; BQ25101 datasheet
U7	Regulator 3.3 V	low-Iq buck/LDO (e.g., TPS62740-class)	—	Iq < 1 µA ideal for sleep	1	<i>unverified</i> — pick in P1
BT1	Battery	Ultra-thin LiPo 3.7 V, 50–120 mAh, protected	402030 (200 mAh, 4 mm thick — too thick for Track A)	thinner = less capacity (VERIFIED: 402030 is 4 mm; a 55 mAh ≤2.5 mm pack exists)	3–5	CM Batteries 55 mAh ≤2.5 mm ; Besomi 402030 ; custom thin cell likely needed
SW1	"Copy" button	low-profile tact switch, ≤1.2	—	IP-sealed preferred	0.3	TBD

Ref	Function	Part (first choice)	Alt	Key spec	Est. \$	Source / link
		mm, side or bezel				
BZ1	Pager alert	piezo SMD buzzer ≤ 1 mm or tiny LRA haptic	—	haptic = "pager-like" feel; thickness check	1–2	TBD
X	Passives, ESD, connectors, antenna keep-out	—	—	—	2–3	PCBWay turnkey/sourced
PCB	4-layer, 0.6–0.8 mm FR-4 (rigid-flex only if camera/display need it)	—	—	ENIG, 40×40 mm outline	3–5	PCBWay <i>unverified</i>

2. Mechanical parts (per unit)

Item	Spec	Est. \$	Notes
Stainless back plate	316L, 40×40 mm, 0.6–0.8 mm, mirror polish	8–20	Polish is the cost driver; quote with finish spec "mirror, Ra target TBD"
Stainless frame/bezel	316L, 40×40 mm, machined window, camera aperture, button hole, vent	15–40	CNC; PCBWay machining mentions 316L work (example) — get quote
Cover glass	0.4–0.5 mm chemically-strengthened, AR coat optional	2–4	
Camera lens window	AR glass or integrated	1	
Gasket / adhesive / micro-screws	PSA, silicone gasket	1–2	
Vent membrane	hydrophobic membrane over 0.8 mm hole	0.5–1	for BME688 airflow

3. Cost roll-up for the 100-unit run (ESTIMATE $\pm 40\%$)

Block	Per unit	×100
Electronic parts + PCB	35–55	3.5–5.5 k
Assembly (PCBWay)	6–15	0.6–1.5 k
Steel + glass + mech	30–70	3–7 k
Per-unit subtotal	~70–140	~7–14 k
One-time: stencil, fixtures, test jig, EVT/DVT rounds	—	3–8 k
One-time: radio compliance (FCC/CE) + UN38.3	—	5–15 k (less if module-based; confirm with lab)
Charger pad (wireless), per unit, any Qi 5 W pad	8–15	0.8–1.5 k

Steel is expected to be the single largest line. A real quote from PCBWay (PCB + assembly + CNC) is a **P2 gate item**.

4. Prototype shopping list for phase P1 (no custom PCB yet)

Order these first (roughly \$150–250, ESTIMATE): 1. ESP32-S3 dev board with PSRAM and DVP camera header (any S3-CAM-style board). 2. 1.3" 240×240 IPS (Adafruit 4520 as known-good reference). 3. BME688 breakout + BMI270 breakout. 4. Qi RX module (Adafruit AF-1901 style, BQ51013B) + 5 W Qi pad. 5. 316L steel

sample squares: 40×40 mm at 0.3 / 0.6 / 0.8 mm (for Qi + RF tests) — local metal shop or PCBWay CNC. 6. 3 candidate thin LiPo cells (see BT1 row). 7. One 3D-printed (resin) 40×40×7 mm shell to check fit.

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LOT-HW1 — Roadmap and Analysis (Document 03 of 06)

Dates are relative weeks from a go decision (W0). Durations are **ESTIMATE**; steel finishing and radio compliance are the long poles. Starting point today (2026-10-04): no hardware exists; the site side already has the pieces to build on (see Doc 05 §1).

1. Phases and gates

Phase	Weeks	Goal	Exit gate (must be true to continue)
P0 Decide	W0–1	S-2 answers the 7 open decisions (Doc 01 §8)	Decisions logged; thickness target set (5 vs ≈7 mm)
P1 Feasibility (dev boards + steel samples)	W1–6	Prove the 3 physics risks on the bench	(a) Qi delivers ≥ 150 mA charge through chosen stack; (b) Wi-Fi link works at ≥ 10 m with steel mock-up; (c) BME688 readable via vent; (d) camera + display + cell fit in a printed shell at target thickness
P2 EVT (10 units, custom PCB)	W6–14	First custom board from PCBWay; steel parts machined	Boots, connects, shows a notification, button writes a Log row; real PCBWay assembly quote in hand
P3 DVT (30 units)	W14–22	Fix EVT defects; power + radio + thermal tests; draft manuals	Battery life measured; drop/dust sanity; compliance lab pre-scan passed
P4 PVT / 100-unit run	W22–32	Production-like build, test jig, packaging, PDF manuals	≥ 95 % first-pass yield on a 10-unit pilot, then release 100
P5 Field	W32+	Hand to first users, collect Log-tab telemetry	Ongoing

Indicative total: 26–34 weeks to 100 units (ESTIMATE). If the 5 mm target is held for Gen-1, add 8–12 weeks (chip-down RF, custom thin cell, thin camera sourcing) and a larger compliance bill.

2. Parallel tracks

Track	Work	Starts
Hardware	schematic → layout → PCBWay package (Doc 06)	P1
Mechanical	CAD of 2-part steel body, polish spec, quotes	P1
Firmware	Doc 04; runs on dev boards in P1	P1
Site/API	Doc 05 endpoints + Log tab event (small, low risk)	P1
Documents	Doc set → PDF manuals; session report each session	P1, continuous
Compliance	lab selection, test plan, UN38.3	P2

3. Critical path

P0 decisions → steel sample Qi/RF tests → PCB layout freeze → PCBWay EVT → DVT → compliance → 100-run

Slack exists on firmware/site work (small); none on steel finishing (long quote/lead) or compliance.

4. Risk register

#	Risk	Likelihood	Impact	Mitigation	Owner
R1	5 mm total thickness not achievable with camera + battery + Qi	High	High	Gen-1 at ≈7 mm; Gen-2 chip-down	S-2 decision
R2	Qi charging weak through steel	High	High	Face-down charging through glass (Option A), ceramic inset (B); test in P1	HW
R3	Wi-Fi range poor in metal body	High	High	Antenna at glass side, keep-out, plastic slot; OTA test in P1	HW
R4	BME688 reads self-heat / no airflow	Med	Med	Vent + membrane, firmware offset, server weather fallback	HW/FW
R5	Steel mirror polish cost/lead too high	Med	Med	Early quote; allow brushed fallback on non-front edges	MECH
R6	Radio compliance cost/time	Med	High	Use pre-certified module in Gen-1	S-2
R7	Thin cell shipping/supply (UN38.3, custom cell MOQ)	Med	Med	Qualify 2 cells in P1; MOQ check	HW
R8	Camera privacy concerns	Med	Med	Capture only on press, visible indicator, no stream (Doc 04 §6)	FW/S-2
R9	Device auth is new surface on the site	Med	High	Per-device tokens, hashed at rest, revocable, rate-limited (Doc 05 §2)	API
R10	PCBWay assembly is manually quoted; part shortages	Med	Med	BOM with approved alternates (Doc 02)	HW

5. Analysis: what to build vs what to buy

- **Buy / off-the-shelf:** ESP32-S3 module, display, camera, BME688, BMI270, Qi RX IC, charger IC. These are the "AI-grade off-the-shelf sensors" you asked for; no custom silicon.
- **Build:** 4-layer PCB, 2-part steel body, firmware, device API on the site, flasher/provisioning tool, manuals.
- **Do not build:** own Qi coil design, own radio, own cell.

6. Scope options for the first 100 (DECISION)

Option	Thickness	Notes	Risk
A "Prove it" (recommended)	≈7 mm	module-based, color IPS, face-down Qi	Low–Med
B "Spec-true"	5 mm	chip-down RF, thin cell, thin camera	High
C "Notify-only"	4–5 mm	drop camera; screen + button + sensors only	Low

Option C is worth a thought: removing the camera removes the biggest thickness driver. If the camera's value is unproven, ship C first and add a camera in Gen-2.

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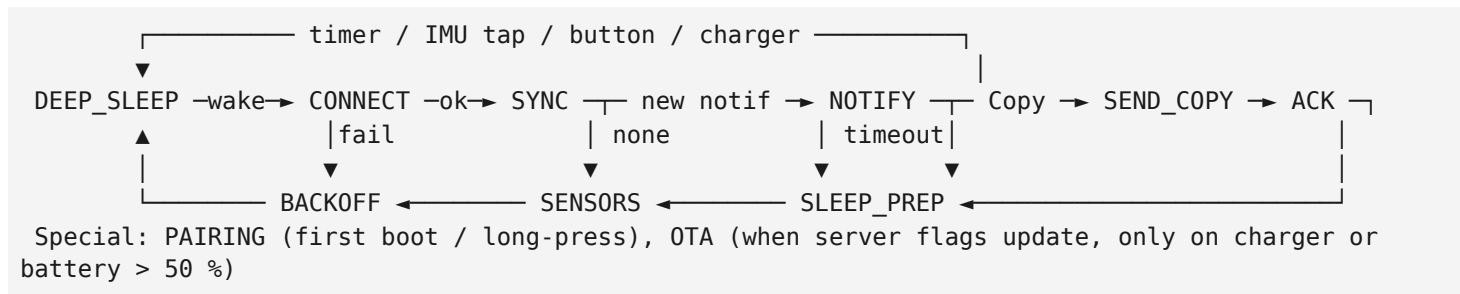
LOT-HW1 — Firmware Document (Document 04 of 06)

Target MCU: ESP32-S3 (dual-core Xtensa LX7, Wi-Fi 4 + BLE, vector/AI instructions — per [Espressif datasheet](#)). Framework: ESP-IDF (recommended for deep-sleep control) with Arduino-compatible only for early prototyping. Status: **design only, no code written**.

1. Responsibilities

1. Wake, join Wi-Fi, fetch pending notifications from the site.
2. Show the notification; vibrate/beep (pager behaviour).
3. On **Copy** press: send a copy event to the site; show confirmation.
4. Read sensors and report them at a low rate.
5. Capture a photo on request, upload, show result.
6. Manage power, charging state, OTA updates, pairing.

2. State machine



3. Timing and power (ESTIMATE; measure in P2)

- Poll interval default 120 s (server-tunable per device, 30 s – 15 min). Backoff doubles on failure up to 15 min.
- Screen on only for NOTIFY (default 20 s) and button feedback.
- Deep sleep current target < 50 µA total board (ESTIMATE; Qi RX and sensors must be quiescent).
- When on a Qi pad (charger present): keep Wi-Fi up, poll every 15 s, screen shows charge state.

4. Display UI

- 240 × 240, brand tokens from the site: background #1a1a1a, accent #43aff3, highlight #fef17b, white text, Arial/Helvetica-class font (VERIFIED in `tailwind.config.js`; re-check against brand site).
- Screens: Notification (text ≤ 120 chars, auto-wrap, 3 lines large), Idle clock + weather glyph, Charging, Pairing code, Error.
- Notification example: *Coffee time!* plus a small footer COPY ▶ log.

5. Input

- One button. Short press = **Copy** (send current notification to Log). Long press (2 s) = dismiss without copy / enter pairing if unpaired. Double press = take photo (Gen-1 option, confirm in Doc 01 §8).
- IMU (BMI270) tap/lift wakes the screen without Wi-Fi.

6. Camera and privacy rules

- Capture **only** after an explicit user action. No streaming, no motion-triggered capture.
- Visible indicator (screen flash/LED-equivalent) for every capture.
- JPEG from the sensor's on-chip encoder; upload over HTTPS; local copy deleted after ACK.
- No microphone in Gen-1.

7. Security

- Per-device token provisioned at pairing; stored in NVS with flash encryption + secure boot enabled on production units (ESP32-S3 supports both).
- TLS 1.2+ to `lot-systems.com` with pinned CA bundle. Reject plain HTTP.
- OTA images signed; rollback on failed boot.
- No credentials in firmware source. Wi-Fi credentials entered via the flasher/provisioning tool (§8) or BLE provisioning.

8. Software that connects to the firmware (the "host" side)

Tool	Purpose	Approach
Web flasher	Flash/update firmware over USB from a browser	WebSerial + esptool-js, hosted as a page on the LOT site or a standalone page
Provisioning	Wi-Fi + pairing	Device shows a 6-digit code; user enters it on <code>lot-systems.com</code> while logged in (Doc 05 §2)
Factory test jig	P4: flash, test sensors/charger/camera/screen/button, print PASS/FAIL, write serial number	Python script over USB serial; logs per-unit results to CSV
Log viewer	Existing Log tab on the site	Shows <code>device_copy</code> entries (Doc 05 §4)

9. Build, test, release

- Repo layout (proposed): `hardware/firmware/` (ESP-IDF project), `hardware/tools/` (flasher, jig), `docs/technical/hardware/`.
- Unit tests on host for protocol parsing and state machine; hardware-in-loop smoke test script for P3.
- Versioning: `HW1-FW-<major>.<minor>.<patch>`; manifest served by site for OTA.
- Each release ships with a changelog and a firmware-doc update (this document) — one source of truth.

LOT-HW1 — LOT API Connector (Document 05 of 06)

Status: **specification only. No server code was changed.** Everything marked EXISTS was read in this repo on 2026-10-04; everything marked NEW is proposed.

1. What already exists on the site (VERIFIED in `src/server/routes/api.ts` and `models`)

Piece	Where	Use for the device
GET <code>/live-message</code>	<code>api.ts</code> ~L820; model <code>LiveMessage</code>	Existing single site-wide message — proves the "site pushes text" pattern, but it is global, not per-device
POST <code>/logs {text, event?, metadata?}</code>	<code>api.ts</code> ~L1563	Creates a Log row for the logged-in user (cookie session). Text is trimmed to <code>MAX_LOG_TEXT_LENGTH</code> .
GET <code>/logs</code>	<code>api.ts</code> ~L1082	Returns logs, filtered by a whitelist of event names (<code>displayableEvents</code>). A new event name will not show in the Log tab until it is added here.
GET <code>/weather</code>	<code>api.ts</code> ~L1038	Server-side weather; usable as fallback for the device
Anthropic client + scheduled jobs	<code>src/server/scheduled-jobs.ts</code> , <code>routes/public-api.ts</code>	AI generation of notification text on a schedule
Auth	<code>routes/auth.ts</code>	Email-code login → cookie session token (<code>Session</code> model). A device cannot use this ; it needs its own token type.

2. Device pairing and auth (NEW)

- Unpaired device calls `POST /api/device/pair/start` → `{pairCode: "483920", deviceId, expiresIn: 600}` and shows the code.
- User, logged in on `lot-systems.com`, opens `Settings` → `Devices` → "Add device" and enters the code → `POST /api/device/pair/claim {pairCode, name}` (cookie-authenticated, existing auth).
- Device polls `GET /api/device/pair/status?deviceId=...` and receives a long random **device token** once. Server stores only a **hash**.
- All later calls: Authorization: `Bearer <deviceToken>`; token scoped to one user and the device endpoints only; revocable from the site; rate-limited (reuse the Fastify rate-limit pattern already used on `/world/generate-element`).

New table `device` (`id`, `userId`, `name`, `tokenHash`, `createdAt`, `lastSeenAt`, `fwVersion`, `revokedAt`). New table `device_notification` (`id`, `userId`, `deviceId?`, `text`, `createdAt`, `deliveredAt`, `source: ai|manual|rule`).

3. Notification flow ("pager-like from an AI site") (NEW)

- A scheduled job (existing pattern in `scheduled-jobs.ts`) or a user rule creates a `device_notification` row, e.g. "Coffee time!". For AI-written text the job calls the existing Anthropic client with a short prompt and a **hard 120-character cap**, content checked server-side before storing.
- `GET /api/device/notifications?since=<id>` → `{items:[{id,text,createdAt}], pollSec}` (ETag/304 supported; empty responses tiny to save power).

- Device marks delivery with `POST /api/device/notifications/:id/ack`.

4. Button "Copy" → Log tab (NEW, small)

- `POST /api/device/events` body `{type:"copy", notificationId, text, ts, fw}`.
- Server validates device token, then creates a Log row for the owning user: `event: 'device_copy', text: <notification text>, metadata: {deviceId, notificationId, source:'LOT-HW1'}` (uses the same `Log.create` as `POST /logs`).
- **Required companion change:** add `'device_copy'` (and later `'device_photo'`, `'device_sensor'`) to the `displayableEvents` list in `GET /logs`, otherwise the row exists but is hidden from the Log tab.
- Idempotency: `(deviceId, notificationId)` unique so a retry after a Wi-Fi drop does not duplicate.

5. Other endpoints (NEW)

Method + path	Purpose
<code>POST /api/device/telemetry</code>	batched BME688/IMU/battery samples (every 10–15 min)
<code>POST /api/device/photo</code>	multipart JPEG ≤ 150 KB; server stores and writes a Log row <code>device_photo</code>
<code>GET /api/device/firmware</code>	OTA manifest <code>{version, url, sha256, signature}</code>
<code>GET /api/device/config</code>	poll interval, brightness, quiet hours

6. Error and abuse handling

- 401 revoked/invalid → device shows "Re-pair" and stops polling.
- 429 → exponential backoff.
- Payload size limits per endpoint; reject unknown type.
- Device events never trigger AI generation directly (prevents loops/cost).

7. Test plan before the first unit exists

- Simulate a device with `curl` / a small script against a local dev server: pair, poll, ack, copy, confirm the Log tab shows the row.
- Unit tests for the new routes (pairing happy path, expired code, revoked token, duplicate copy).

LOT-HW1 — Manufacturing and PDF Manuals

(Document 06 of 06)

1. PCBWay order package (for EVT, repeated for 100-run)

PCBWay assembly is quoted manually after file review (typically 1–2 business days, per research; vendor page itself could not be opened from the sandbox — confirm at pcbway.com/pcb-assembly).

Deliverable	Content
Gerber + drill	4-layer, 40×40 mm outline, 0.6–0.8 mm, ENIG
BOM	LOT - HW1 - BOM . csv extended with manufacturer part numbers and approved alternates
Pick-and-place	centroid file from the EDA tool
Assembly drawing	polarity, keep-outs (antenna, Qi coil), no-fill areas
Stack-up note	impedance for RF trace, thickness
Test notes	5 test points: 3V3, GND, BAT, VQI, UART
Quantity plan	EVT 10, DVT 30, PVT/run 100 (+10 % spares ESTIMATE)

Decision: **turnkey** (PCBWay sources parts) for production; **consigned** for scarce parts (BME688, display) if lead time bites.

2. Steel body package

- 3D STEP files for both parts; drawing with tolerance ± 0.05 mm on mating faces (ESTIMATE; confirm with shop).
- Material 316L, finish: back plate mirror polish (target spec to confirm), frame bead-blast or satin (DECISION).
- Ask for: sample set of 3 finishes, per-unit price at 10/30/100, lead time, inspection report.
- Candidate source: PCBWay CNC machining ([example project](#) shows 316L work); get at least one local alternate quote.

3. Assembly and test flow (100-run)

1. PCB assembled by PCBWay → incoming inspection (visual + continuity on 10 %).
2. Flash firmware + write serial number via jig (Doc 04 §8).
3. Functional test: display colours, button, camera capture, BME688/BMI270 read, charge-in via Qi, Wi-Fi + site round trip.
4. Cell attach, glue, seal, final assembly, leak/vent check.
5. Burn-in 24 h on charger, then PASS/FAIL label and QR (serial → device record).
6. Pack with Qi pad, quick-start card, safety sheet (lithium cell).

4. PDF manuals set (item 7)

These are generated **from the Markdown sources in this folder** so the PDFs never drift from the docs.

PDF	Audience	Source
LOT-HW1 Engineering Pack	internal	Docs 01–06
LOT-HW1 Firmware Manual	firmware engineers	Doc 04
LOT-HW1 API Connector Guide	site/backend	Doc 05
LOT-HW1 Quick Start (1 page)	end users	written in P4
LOT-HW1 Safety & Compliance sheet	end users / lab	written in P3–P4

Build command (this session produced the engineering pack): `docs/technical/hardware/build-pdf.py` renders the Markdown files to `LOT-HW1-ENGINEERING-PACK.pdf` with headless Chromium.

5. Compression of information each session (item 8)

Each working session ends with one session report (`docs/benchmark/LOT-SR-YYYYMMDD-NN.md`), one ledger line, and lexicon/doctrine updates, per the LOT benchmark protocol. Hardware status lives in this folder; the report only records what changed.

6. Compliance checklist (start in P3)

- FCC Part 15 (US), CE-RED (EU) if shipped there — module-based Gen-1 reduces scope; lab confirmation required.
- UN38.3 for the lithium cell; battery protection circuit documented.
- Qi: interoperability tests on ≥ 3 pads; no Qi logo claim.
- Camera: privacy statement in the Quick Start; capture indicator documented.
- Marking: "Made in the USA" claim (used on `docs/README.md`) must match the actual build location/content — confirm before printing on the product.

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